BRICS

Brasil 2025

SUL GLOBAL INCLUSIVO E SUSTENTÁVEL

BRICS
Brasil 2025
COOPERATING FOR AN INCLUSIVE AND SUSTAINABLE WORLD

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FINAL AGENDA

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[SATURDAY], [MAY] [17], 2025

[09:00 AM] [Session 1 - Welcome remarks and general information]

[09:05 AM] [Minister Luciana Santos' video and Vice Mayor of Petropolis, Albano Baninho]
[09:30 AM] [Speech of the representative of Brazil]
[09:35 AM] [Speech of the representative of China]
[09:40 AM] [Speech of the representative of India]
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[09:50 AM] [Speech of the representative of South Africa]
[09:55 AM] [Speech of the representative of United Arab Emirates]

[10:00 AM] [Group Photo & Coffee Break]

[10:20 PM] [Session 2 - The BRICS opportunities for further collaboration]

[10:25 PM] [Federated Social Networks for BRICS]
[10:45 PM] [Storage for BRICS Cloud]
[11:05 PM] [Santos Dumont Supercomputer Ambassadors Program]

[11:25 AM] [Session 3 - Overview of BRICS ICT-HPC Landscape]

Presentations by BRICS countries on areas of interests from ICT and HPC with potential collaboration topics.
[11:25 AM] [Brazil]

[11:45 PM] [Lunch Break]

[01:45 PM] [Session 3 - Overview of BRICS ICT-HPC Landscape (continued)]

[01:45 PM] [Russia]
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[04:05 PM] [Coffee Break]

[04:25 PM] Delegates will undertake a technical visit to the National Laboratory for Scientific Computing (LNCC) in Petrópolis, Rio de Janeiro, to gain insights into the Supercomputer Santos Dumont, the most advanced academic supercomputer in Latin America since 2015.

[07:00 PM] Dinner

DAY 2

[SUNDAY], [MAY] [18], 2025

[10:00 AM] [Session 4 - Update on Flagship projects]

[10:05 AM] [“Digital Earth” Priority Flagship Project]
[10:35 AM] [Presentations by national teams (update on the progress in national activities related to “Digital Earth” project) (Russia, South Africa, Brazil, China, India)]

[11:00 AM] [Coffee Break]

[11:20 AM] [Flagship Project on AI for Education (China)]

[01:00 PM] [Lunch Break]

[03:00 PM] [Session 5 - Discussion on thematic areas for BRICS STI Programme Calls]

[03:00 PM] [(Flagship and regular projects) and other modes of collaboration]
[03:30 PM] [Discussion]

[04:00 PM] [Session 6 - Minutes and Meeting Closure]

Presentation of Minutes structure, discussion, final remarks by countries and conclusion
[04:45 PM] [United Arab Emirates]
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Minutes

BRICS LNCC Petrópolis 2025

Day 1

[Session 1 - Welcome remarks and general information]

Representatives of the BRICS countries got together in Petrópolis/RJ, on May 17th and 18th, at the Headquarters of the National Laboratory for Scientific Computing (Laboratório Nacional de Computação Científica – LNCC), for the 9th Meeting of the BRICS Working Group on High-Performance Computing (HPC) and Artificial Intelligence (AI).
The Meeting was opened with a video message from the Ministry of Science, Technology, and Innovation, Luciana Santos. In the sequence, the Vice Mayor of Petropolis, Albano Baninho, delivered a welcome to the Delegates. Finally, the Delegates representing the countries made welcome remarks in the order, Brazil, China, India, Russia, South Africa, and United Arab Emirates. Respectively, the Delegates were Fábio Borges de Oliveira (LNCC), Aimin Zhou (Dean of Shanghai Institute of Artificial Intelligence for Education), Sibashisa Dash (Department of Science and Technology, Government of India), Viktor Stepanenko (Lomonosov Moscow State University), Happy Marumo Sithole (CSIR), Saeed Aljaberi (TII).

[Session 2 - The BRICS opportunities for further collaboration]

In accordance with the 2025 Brazil's leadership guidelines, representatives of Brazil proposed 3 initiatives for further discussion by the BRICS countries:

- The Federated Social Network for BRICS, where BRICS countries could have their own digital public space to share ideas, without dependencies of non-BRICS platforms, to strengthen the cooperation among Global-South countries.
- The Storage for BRICS Cloud, an initiative that aims to create a model for collaborative and voluntary data sharing that could preserve BRICS countries national heritages, enable AI training that better reflect the BRICS countries cultures and

- The Santos Dumont Supercomputer Ambassadors Program, a collaboration program to match a Brazilian institution and a BRICS country institution into a joint research and use the Santos Dumont Supercomputer on the research.

  Details of this section proposals are in the appendix.

[Session 3 - Overview of BRICS ICT-HPC Landscape]

On the overview of the BRICS ICT-HPC landscape, the representative of the countries presented as follows:

Brazil presented the Brazilian AI landscape (the AI research centers, the AI related publications, the main research areas and collaborations), the Brazilian Plan for AI – PBIA (amount of investment, the 5 axes) and the HPC-AI research landscape: the Brazilian approach, the RISC-V collaboration, memory-based technologies and HPC-AI applications, including the following research domains: Bioinformatics, Health, Geophysics, Climate Change, Large Language Models and HPC in Clouds. Brazil showed research results in all those domains from groups led by researchers from several universities (UnB, UFF, UFRGS, UFMG, UFSC, UFSCAR, UNICAMP) and research institutes (LNCC, ITA and INPE).

Russia presented the Lomonosov-2 supercomputer and the potential collaboration topics in HPC-AI (performance analysis and optimization of applications, development of personalized medicine, drug design and development, natural language processing, adaptation of large language models to Russian, optimization of LLMs, and foundation modeling for climate science).

India presented the National Supercomputing Mission (creation of capacity and capability with the development of indigenous technologies and human resources) and the areas of interest (HPC applications for grand challenge problems; genomics and drug discovery; weather, air quality and climate modeling; natural hazards mitigation).

China presented a report highlighting three key developments on HPC-AI from China. First, China leads in HPC with over 100 systems on the TOP500 list and three major vendor types, while mixed-precision computing emerges as a trend due to AI's impact. Second, Shanghai Jiao Tong University's (SJTU's) HPC Center, with 30+ years of expertise, operates advanced platforms like the Siyuan-1 supercomputer, featuring 90,000+ CPU cores and 1,000+ AI cards, supporting top-tier academic research. Third, global collaborations are driving innovation, including the newly established Global Computing Consortium (GCC), which promotes Arm-based HPC through events like the ABS4S workshop and open-source projects. Additionally, SJTU recently won the ASC25 championship in the ASC Student Challenge, showcasing its commitment to fostering HPC talent. This report underscores China's leadership, SJTU's contributions, and the importance of international collaboration in advancing HPC technologies.

South Africa presented the National Integrated Cyber Infrastructure System - NICIS, the only HPC for research provider in South Africa, and its platforms and utilization, as well as its specialized HPC computing services. As areas of collaboration, South Africa pointed to HPC and AI applications, OpenStack Cloud development, Quantum Computing, and community events.
The United Arab Emirates presented the National Strategy and Vision (vision 2031), the HPC infrastructure (including the Jubail supercomputer, Ankabut HPC Platform, UAEU HPC Infrastructure), cloud HPC integration, HPC applications (weather and climate modeling, oil and gas, smart infrastructure) and international collaboration and outreach (Atlas Cluster and CERN, G42 and VAST Data, Technology Innovation Institute). UAE also presented its challenges and future vision.

Day 2

[Session 4 - Update on Flagship projects]

Digital Earth Flagship project

Dr. Viktor Stepanenko (Russia) presented an overview and updates on the priority flagship project «Digital Earth». He reminded that weather and climate extremes are widely recognized as the top jeopardies for BRICS countries, which is underlined by the recent BRICS Joint Declaration for reduction of climate disasters (9 May 2025, Brazil). The goal of the project is to create efficient cutting-edge indigenous HPC (physical)+AI models facilitating the prediction of extreme weather/climate conditions with focus on highly urbanized areas of BRICS countries, quantification the impacts of those conditions on socio-economic sectors, and provision of products in demand for the general public and stakeholders. It pursues the federated development paradigm across BRICS nations which leads to mutual beneficiation and human capacity growth in the subject area. This paradigm is further detailed in the project as tool-to-tool and tool-to-use-case interactions. HPC and AI prediction tools and hotspot use-cases (urban areas, suffering from floods, droughts and harmful air quality) are identified for each country. Work packages and project timeline are updated.

Professor Eduardo Bezerra presented Brazil's contributions to Digital Earth, highlighting the RioNowcast project and the Gypscie platform. RioNowcast integrates diverse meteorological data sources—radars, satellites, stations, and models—into a unified infrastructure for training AI models focused on extreme rainfall nowcasting. The Gypscie platform supports the full AI model lifecycle, from training to deployment and monitoring, enabling model reproducibility and sharing. Both tools are aligned with Digital Earth objectives and promote interoperability through common data structures and APIs. Bezerra proposed collaborative model comparisons and infrastructure sharing among BRICS nations. The initiatives also align with Brazil's AI Strategy (PBIA), emphasizing technological sovereignty, renewable energy, and climate resilience. Brazil offers expertise in deep learning, big data, and AI education, and its tools are ready for international collaboration.

Dr. Jinxi Li presented Chinese background and contribution to Digital Earth. Institute of Atmospheric Physics CAS develops unique multiscale (regional-to-local) non-hydrostatic adaptive-grid modeling system Fluidity with primary applications in heavy rainfall and urban air quality prediction. The system effectively uses modern CPU and GPU architectures, as well as AI for governing the mesh resolution. China proposes tool-to-tool interactions (Fluidity to Russian MSU-INM LES model intercomparison, Fluidity to Russian TerM model coupling, AI methods for adaptive grid with Brazil, sharing HPC resources for installation and efficiency studies) as well as tool-to-use case cooperation (urban scale modeling for selected largest BRICS cities).

India presented two technologies to be shared with BRICS partners and adapted to hotspot use-cases: (i) High-resolution HPC-based flood forecasting and early warning system for Large river basin across BRICS Nations, and (ii) a High-resolution (down to building-scale) HPC-based Integrated urban air quality, weather (extreme rainfall, temp, etc,) and flood forecasting system for urban cities across BRICS nations. The technical parameters of input (initial, boundary conditions, land surface characteristics) and validation data, as well as overall dataflow and final forecasting products for decision-makers are specified. The Indian group is led by Prof. Manoj Khare (C-DAC).

Russia (Dr. Viktor Stepanenko) presented an indigenously developed toolkit for modeling and prediction of weather and climate dynamics from global to local scales. It includes physics-based models — global Earth system model INMCM, participating in CMIPx experiments, land surface model TerM, regional-to-local modeling system MSU-INM, and a number of AI-facilitated models for downscaling of coarse-resolution hydrodynamical forecast. Protocol of the intermodel comparison of MSU-INM model with Fluidity is sketched. First results of application of TerM model to Limpopo river use-case in South Africa are demonstrated as an example of tool-to-use case collaboration.

South Africa (Dr. Happy Sithole) provided an overview of the weather and climate modeling landscape in the country as well as recent advances (including the Limpopo use-case for the Russian TerM model) and suggestions for further collaboration in the Digital Earth framework. The latter include application of the

Russian SLAV model and Chinese Fluidity model to regional-to-local forecasting over South Africa. Priority use-cases for prediction tools were identified as Durban, Gauteng and Tugela river.

AI for Education Flagship project

Dr. Aimin Zhou from Shanghai Institute of AI for Education (SIAIE), East China Normal University, updated the recent work on the candidate flagship project on AI for education. As one of the most important issues raised at the Working Group meeting, Artificial Intelligence (AI) was highlighted as a catalyst for educational reform. AI empowers educational innovation and promotes educational development to advance quality-oriented education. Dr. Zhou provided an update on SIAIE's recent research projects in AI for education, including large language models (LLMs) for education and multiple LLM-based applications. Supported by AI technologies, two potential collaboration opportunities were proposed: Research on LLM for education tailored for diverse education systems (e.g., specialized applications in K-12 mathematics education); Research on enhancing LLM reasoning capabilities through the design of complex mathematical problems.

[Session 5 - Discussion on thematic areas for BRICS STI Programme Calls]

Considering that all participating countries have agreed to pursue the flagship project on "Digital Earth", members from all countries unanimously agreed that this decision be brought to the notice of the funding agencies of respective countries, for them to enable approving this project.
Guidance to be sought from the funding agencies, and the BRICS Funders Working Group to decide how the flagship project can be pursued further leading to its approval.

1. **Federated Networks for Social Media**
   The proposal is to use a common BRICS platform for social media, which is independent from the other platforms. There are already some countries that have platforms, such as China with WeChat, etc. The proposal is to set-up a technical committee that can explore the possibility of implementing such a platform and report to the Working Group in the next meeting.

2. **BRICS Storage and Data Sharing**
   The proposal was made by Brazil to consider for the Working Group Members to have a shared storage for the BRICS member countries. The proposal is supported by all the member states. The member states further propose that this be in the same design as the BRICS Hub, and all the countries that propose a Technical

Committee to discuss the requirements to provide resources for this purpose. A possibility the members propose is that Flagship Projects Digital Earth can be used as a Proof of Concept for data storage and sharing.

3. **BRAZIL Ambassador Program on HPC resources**
Brazil has proposed to develop an Ambassador Program that will allow other researchers from other countries to have access to Santos Dumont. The condition for accessing the resources will be for a PI to be in Brazil and other researchers have a collaboration with Brazil PI. The proposal is supported by Working Group Members, and this will enhance the objectives of the WG of collaboration amongst the member countries. It is further suggested that this proposal be extended by all the Member Countries, to make available resources in their countries with the same conditions as those proposed by Brazil.

The group recommended that a quantum computing session should be included at the meeting in 2026.

We look forward to the next meeting, which probably should take place in India, as the Working Group will follow the host country in 2026.

[Session 6 - Minutes and Meeting Closure]

The countries Iran, South Africa, China, India, Russia, Brazil made the meeting closure by the delegates Reza Rasoolzadeh, Tebogo Mokoma (Director: Overseas Bilateral Cooperation), Aimin Zhou (Shanghai Institute of Artificial Intelligence for Education), Ashish Pandurang Kuvelkar (C-DAC), Viktor Stepanenko (Lomonosov Moscow State University), Fábio Borges de Oliveira (LNCC).

Delegate List

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Appendices

Federated Social Networks for BRICS

In light of the growing prominence of the BRICS on the international stage and the need to strengthen sovereign, inclusive, and culturally sensitive communication channels — and in line with the guidelines of Brazil's presidency in 2025, which aim to reinforce the strength and cohesion of the bloc — we present the proposal for the creation of a Federated Social Network of the BRICS.

This initiative aims to foster a collaborative and decentralized digital space among member countries, using open, established technologies and integrating artificial intelligence (AI) resources for translation, moderation, multilingual content analysis, and sentiment analysis. Authentication may be carried out using national identification documents or existing services such as Brazil's gov.br, ensuring security and verified identity.

The federated model allows each member country to maintain its own instance of the social network, with autonomy to define its rules for moderation, data storage, and governance, while remaining interconnected with the other instances of the bloc. This ensures interoperability among BRICS users while respecting the political, cultural, and legal particularities of each nation.

Additionally, member countries will have the freedom to join the project gradually or partially, in line with their priorities and capabilities. In this way, we aim to provide a respectful, safe, and constructive digital environment for all participants of the bloc.

Main Objectives:

Strengthen the digital sovereignty of BRICS countries by promoting secure, open, and decentralized technological alternatives.

Enhance communication between citizens, institutions, and governments of the BRICS through decentralized digital platforms that value the bloc's cultural diversity.

Promote content moderation based on each country's laws, cultures, and values, respecting national sovereignty and avoiding the imposition of external standards.

Encourage horizontal cultural, scientific, and educational exchange, with digital spaces for academic, artistic, and social collaboration among the peoples of the BRICS.

Stimulate regional technological innovation, encouraging participation from universities, startups, and open-source software communities in developing the network.

Create a federated and interoperable ecosystem, capable of integrating different digital services, such as social networks, messaging systems, and marketplaces—in a secure and scalable manner.
Promote digital inclusion and civic participation, with accessible, multilingual tools adapted to the diverse social and economic realities of the bloc's countries.

Expected Benefits:

Greater integration among the peoples of the BRICS, strengthening intercultural dialogue and a shared digital identity.
Stimulation of local innovation and international cooperation, driving scientific and technological development in member countries.
Creation of a resilient and autonomous digital ecosystem, with reduced dependency on external infrastructure and greater control over data and algorithms.
Equitable digital inclusion, respecting the linguistic, economic, and cultural diversity of the bloc's populations.
Support for regional entrepreneurship, with the creation of a federated marketplace for local products and services in member countries.
Encouragement of citizen participation and transparency, through accessible tools for public consultation and collective decision-making.
Dissemination of best practices, allowing successful solutions to be replicated across countries while respecting local specificities.
Synergy with other strategic BRICS initiatives, such as academic networks, joint laboratories, and media observatories.
This proposal was presented during the 9th Meeting of the BRICS Working Group on High-Performance Computing and AI, which will take place on May 17–18, 2025, at the National Laboratory for Scientific Computing (LNCC) in Petrópolis, RJ, Brazil. The initiative is expected to contribute to deeper digital cooperation among the bloc's countries, paving the way for a new paradigm of digital sovereignty, technological innovation, and integration among the BRICS peoples.

BRICS Model for Collaborative Data Storage and Sharing

Overview

This proposal establishes a collaborative framework for data storage and sharing among BRICS countries, aiming to preserve and leverage their members' rich cultural, scientific, and educational heritage. The model is built on creating national, redundant data clouds in each country, where public domain and non-public data are securely stored and made available for controlled, sovereign, and transparent sharing. The initiative is designed not only to safeguard vital assets from loss due to disasters or conflicts but also to foster technological advancement, particularly in artificial intelligence, by enabling the training of models that reflect the diversity and realities of BRICS societies.

Principles

At its core, the framework prioritizes the protection and accessibility of public and semi-public data, such as theses, dissertations, scientific articles, museum collections, and linguistic records, including those of indigenous and minority languages, which are currently dispersed or at risk. Each country maintains local repositories with replicated data, ensuring geographic redundancy and resilience. Data is stored in high resolution and with robust security, supporting the preservation of valuable resources like genetic data from archaeological findings. Non-public data, including copyrighted or sensitive materials, is held with restricted access, employing federated learning and encryption for privacy and compliance. Such data remains protected until it transitions to the public domain, at which point it becomes accessible and preserved within the national clouds. The proposal also recommends that national legislation be updated to allow for early storage and timely public release, ensuring that data is maintained and available for scientific and cultural use as soon as it enters the public domain. The framework draws on international best practices for voluntary, non-binding data sharing that respects digital sovereignty and national laws, promoting regulatory convergence and open-by-default policies.

Technical Architecture

The technical foundation is a federated repository structure, with each country maintaining national repositories aligned with the FAIR Principles (Findable, Accessible, Interoperable, Reusable). Interoperability is achieved through APIs that enable cross-system queries and granular access control by project or thematic domain. Data centers equipped with high-performance hardware and dedicated networks ensure redundancy and physical security, and the countries are interconnected by a dedicated fiber-optic submarine cable network, reducing reliance on external infrastructure and enabling secure, high-speed data transfer. The repositories are organized by thematic domains, such as cultural history, applied science, or languages, with adaptable permissions for specific projects, allowing, for example, an Indian researcher to access Brazilian climate data or a Russian museum to share digital artifacts with South Africa, all while maintaining sovereignty over the original data.

Governance and Standards

Governance is based on multilateral agreements that respect national laws and do not impose mandatory sharing, instead incentivizing voluntary participation and mutual contribution. Each nation determines what data to share and defines access policies for different user groups using role-based access control, with hierarchical levels such as public, restricted, or confidential. Automated provenance tracking records the origin, modifications, and transfers of data, integrating audit metadata to ensure compliance with ethical and legal standards. Dispute resolution is facilitated through blockchain-based smart contracts, automating licensing and rights management. The model recommends harmonizing national copyright laws for early preservation and timed public data release, ensuring the framework remains adaptable and legally sound.

Use Cases

The framework supports a range of immediate and future use cases. These include the creation of unified repositories for indigenous and minority languages, the integration of climate and environmental data for predictive modeling in agriculture and disaster response, the coordinated digitization of government archives for comparative historical research, and the collaborative training of AI models on local datasets to reduce cultural bias and enhance representativeness. In later phases, sensitive domains such as health data may be included, using advanced cryptography and federated learning to ensure privacy and compliance.

References and Existing Models

Several existing initiatives provide technical and conceptual foundations for this framework. CyberBRICS offers governance and legal strategies for digital sovereignty and cybersecurity. CBAS Data Products demonstrates thematic interoperability and open formats for geospatial data sharing. ScienceDB, maintained by the Chinese Academy of Sciences, provides a model for flexible licensing, FAIR-compliant curation, and integration with analytical tools. The Brazilian Gypscie framework enables privacy-preserving AI training without raw data transfer, supporting legal compliance and resource optimization. These models illustrate how solutions developed within the BRICS context can sustain a federated architecture that preserves sovereignty while enabling cooperation.

Benefits

The collaborative framework delivers substantial benefits to all BRICS members, including preserving and controlling national and collective heritage, enhanced research capacity and innovation, increased resilience through redundancy and collaboration, and the opportunity to shape international digital standards and practices. By enabling the training of AI models on diverse and representative datasets, the framework also addresses cultural bias and promotes fairness in emerging technologies.

Conclusion

This BRICS Collaborative Data Storage and Sharing Framework proposal offers a robust, adaptable, and forward-looking approach to digital sovereignty, heritage preservation, and technological development. It addresses the technical, legal, and political challenges inherent in a multilateral initiative of this scale, aiming to strengthen the collective capacity of BRICS countries to safeguard their assets, foster innovation, and lead in the global digital landscape.

Santos Dumont Supercomputer Ambassadors Program

Dr. Carla Osthoff from LNCC presented an overview of the institution and its strategic initiatives. She began by introducing LNCC (National Laboratory for Scientific Computing) and providing a general overview of the research institute. She then highlighted LNCC's Artificial Intelligence research groups and emphasized Brazil's participation in the "Digital Earth" initiative, introducing Prof. Eduardo Bezerra from CEFET-RJ as the Brazilian representative in this collaboration.

Following this, Dr. Osthoff presented LNCC's High-Performance Computing (HPC) research group, outlining their involvement in the national AI software stack project and the RISC-V educational initiative, both part of Brazil's National Plan for Artificial Intelligence (PBIA).

She also detailed the architecture of the Santos Dumont supercomputer, the range of research areas supported by its users, and production statistics. This context served as the basis for introducing the **BRICS Ambassador Program**.

Dr. Osthoff explained that each BRICS country is invited to nominate an **Ambassador**—a representative who will foster collaborative research between their home country and Brazil. The Ambassador's role is to identify research groups in their country that are interested in collaboration and have computational problems that require supercomputing resources. LNCC will support this process by helping match them with appropriate Brazilian research groups if needed.

To be eligible, the Brazilian coordinator of the joint project must be affiliated with a recognized educational or research institution in Brazil. LNCC will facilitate the collaboration, providing support such as arranging meetings, offering participation in the HPC Summer School, and providing training opportunities.

Interested representatives can contact LNCC at **diretoria@lncc.br** for more information or to initiate the collaboration process.

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Proposals on AI4Education

Project 1. Construction of a Large Language Model for K12 Mathematics Education: Building a LLM and agents tailored for the K12 education to support student development.

Research Contents:

(1) Collection of mathematics education-related corpora, including a complete set of K12 mathematics textbooks, mathematics teaching syllabi, well-covered mathematics exercises with correct answers, students' answers to the exercises, and dialogue corpora during mathematics teaching and learning.

(2) Annotation of mathematics problems and data from mathematics teaching and learning process, including the knowledge points corresponding to mathematics problems, the difficulty levels of mathematics problems, annotations of students' answers, annotations of teaching dialogues, etc.

(3) Training of a LLM for mathematical education, utilizing the above corpora to train the model.

(4) Development of applications based on the LLM for mathematical education, focusing on building various intelligent teaching agents. These include agents that support teachers in generating and grading math problems, as well as agents that assist students in explaining and analyzing mathematical problems.

(5) Applied research on the LLM for mathematical education, investigating the impact of the LLM and intelligent agents on education within specific teaching processes, and conducting international comparative studies.

Project 2. Training of a Large Reasoning Model: Enhancing the reasoning capabilities of a large model by constructing challenging mathematics problems.

Research Contents:

(1) Design of challenging mathematics problems: generating problems that exceed the difficulty level of Mathematical Olympiad problems.

(2) Training of the large model: utilizing challenging problems to train the reasoning capabilities of the large model.

(3) Organizing human-machine hybrid international mathematics competitions: jointly organizing human-machine hybrid mathematics competitions and releasing rankings of the large model's problem-solving abilities in mathematics.

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Digital Earth Flagship Project Concept

Common Challenges of the BRICS countries. World Economic Forum (2023) identifies a failure to adapt to climate change and extreme weather events (such as heatwaves, heavy precipitation, floods, droughts, wildfires, bad air quality) as the top global risks over a horizon of 10 years. To address these issues, consortia of developed countries launch ambitious projects for creating digital twins of the Earth system, e.g., EU Earth digital twin project. BRICS countries in particular suffer from weather and climate extremes, reliable prediction of which is of vital importance for sustainable development of BRICS societies, especially in highly urbanized areas. Thus, the long-term partnership between BRICS countries in developing national cutting-edge Earth system digital twins is vital in order to minimize the hazardous impacts of extremes on natural economies and society as well as to better plan the green transition and build-up constructive BRICS position at climate-related international affairs.

Scope of Project. Combination of advanced physics-based weather and climate forecast technologies and artificial intelligence is a cutting-edge paradigm for improvement of natural extremes prediction. Each BRICS country develops indigenous multiscale full-chain weather and climate prediction systems moving in this concept. The flagship project aims to enhance the predictive capabilities of national systems by interchange of methods, programming codes, data, and expertise between BRICS partners via common R&D ecosystems in order to improve each country's forecast technological lines. The differences between current national technologies will be systematically treated as complementaries, where any country will be able to provide the "bricks" to another partner's system. Given the variety of weather and climate codes and HPC systems developed and operated in BRICS countries, systematic study will be performed to quantify the extent of conformity between weather forecast algorithms and a wide range of hybrid HPC configurations available to BRICS partners. The co-design of software and HPC architectures will be performed to qualitatively improve the efficiency of weather and climate codes. The projects will adopt the innovative coupling of Earth system models and the models of anthropogenic systems in order to explicitly reproduce the socio-economic impacts of weather and climate extremes and feedbacks between natural and human systems. The socio-economic models will include energy, agriculture, and demographic sectors. The priority extremes will include precipitation, floods, droughts, wildfires, air quality and meteorological comfort with focus on the largest megacities of BRICS countries. The forecast products and operational services will meet the key demands of each country's citizens, stakeholders, and governments.

Interdisciplinary nature. The project is highly interdisciplinary. It involves research institutions, universities, national weather services and HPC centers from all BRICS countries. The expertise of partners includes data-driven (AI) models and large-scale HPC systems management, atmospheric boundary layer modeling, urban meteorology and land surface schemes, fine scale urban flood and atmospheric precipitation extremes forecast, fine-scale river-basin flood prediction and forest fire spread simulation, multiscale numerical weather prediction model development, data assimilation and AI-parameterizations for subgrid processes. The complementary backgrounds of project teams will ensure the competence transfer between partners to ease the solution of country-specific problems of environmental prediction.

Novelty. The main scientific innovativeness of the project is the creation of tools for efficient prediction of extreme events and conditions in the Earth system, based on the integrated use of mechanistic and data-driven modeling techniques targeting optimization of HPC resources usage, forecast accuracy for most hazardous extremes, and matching the demands of weather-vulnerable socio-economic sectors.

BRICS uniqueness.

- The new collaboration strategy based on the equitable and constructive exchange of products and expertise using joint R&D ecosystems, which is inherent to BRICS spirit is implemented in the area crucial for all partners and the world (weather and climate prediction);
- The principles of supercomputer codesign are first systematically applied to a particular subject area;
- The coupling of models of the Earth system and socio-economic sectors for comprehensive account for impacts and feedbacks between weather/climate and humans;
- The weather/climate/environment national prediction systems are improved to address the country-specific natural risks.

Scientific and Socio-Economic impact. There is a vital demand for accurate and reliable information on extreme weather conditions in cities, floods, droughts and wildfires, for appropriate action and timely warnings to the community, stakeholders and governments. The provision of prediction products and services adapted to demands of country-specific consumers will ensure a direct positive economic effect. The technologies to be created in the project will be integrated with the information dissemination systems for quick dispatch of information to disaster response units. Project will create new research infrastructure for scientists from BRICS countries dealing with fundamental and operational problems of Earth system predictions. This will foster the resilience of the urban infrastructure and landscapes and mitigate the unfavorable climate change and natural disasters across all countries. Apart from scientific and technological output of the project, coordinated training courses will be organized to teach students, scientists and operational forecasters the new models and techniques developed in the project.